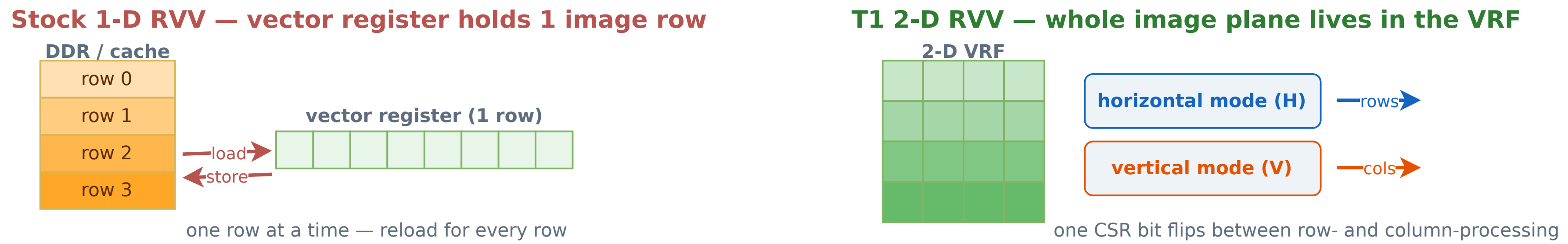
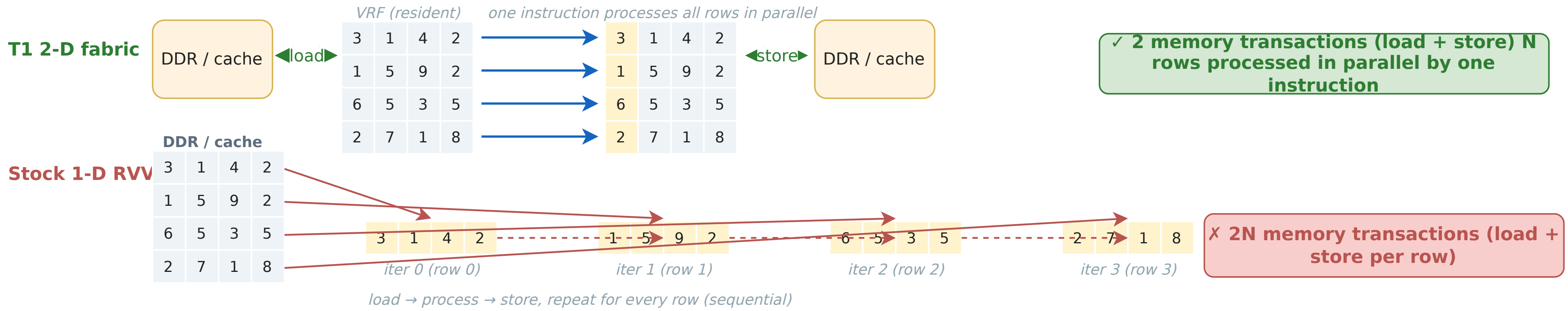
Stock 1D RVV vs T1 2D RVV: data residency & H/V mode flip

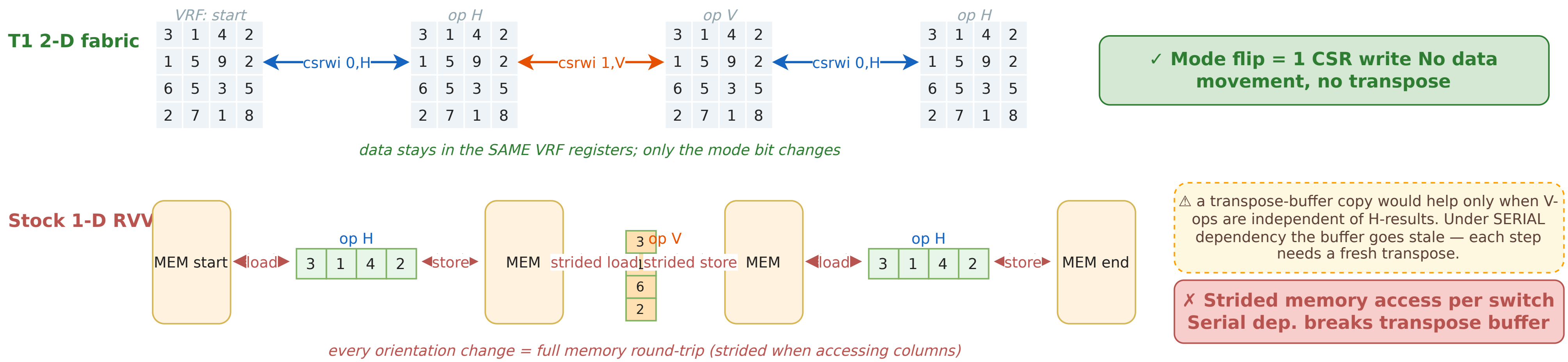
On a 1D vector machine each image row must be fetched into a register and stored back; switching orientation forces strided memory access.
T1 keeps the whole plane in the VRF and toggles horizontal↔vertical with one CSR bit.



Task A · Process all N rows of an image (e.g., a per-row horizontal pass)



Task B · Alternating Horizontal & Vertical operations (serial data dependency)



Summary

Aspect	Stock 1D RVV	T1 2-D fabric
Image residency	streams row-by-row through registers	whole image plane lives in VRF
Memory traffic (N×N image)	2N transactions (load + store per row)	2 transactions (load once, store once)
Switching H ↔ V	strided memory load + strided store	one CSR bit (csrwi 0x7c0)
Serial dependent H/V chain	each step pays the full memory round-trip; transpose-buffer cannot be reused	data never leaves VRF; mode flips between consecutive instructions